Claims

1

A human-machine collaboration system based on standardized cognitive protocols, characterized in that the system comprises:
a standardized cognitive protocol container, including a metadata header, an exclusive mantra parameter mapping set, and a Physical Sincerity assertion predicate package; wherein the metadata header contains a globally unique hash primary key representing the current protocol and a globally unique hash fingerprint of a parent protocol;
a meta-cognitive protocol compiler deployed in a secure computing environment and connected to the standardized cognitive protocol container, configured to detect characteristics of a target computing power engine, and dynamically compile and assemble the exclusive mantra parameter mapping set and the Physical Sincerity assertion predicate package into a translated instruction stream adapted to the characteristics of the target computing power engine;
and an outbound self-consistency assertion interceptor deployed in the secure computing environment, configured to intercept raw output streams returned by the target computing power engine, release the raw output streams when the raw output streams satisfy judgment conditions specified in the Physical Sincerity assertion predicate package, and block the raw output streams otherwise.

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The human-machine collaboration system according to Claim 1, characterized in that the globally unique hash primary key in the standardized cognitive protocol container is generated through cascading calculation as follows:
performing standardized serialization processing on a combined data body of the exclusive mantra parameter mapping set and the Physical Sincerity assertion predicate package to generate an ontology data stream, splicing the ontology data stream with the globally unique hash fingerprint of the parent protocol, and executing a cryptographic SHA-256 hash operation; for an original root protocol, a preset 32-byte all-zero data is filled into the field of the globally unique hash fingerprint of the parent protocol.

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The human-machine collaboration system according to Claim 1, characterized in that the metadata header further comprises:
a context Schema verification field uniformly pointing to a standard definition namespace;
a session-level lifecycle time threshold **TTL**;
and an asymmetric anti-counterfeiting digital signature generated by encrypting the entire container payload with a publisher's private key.

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The human-machine collaboration system according to Claim 1, characterized in that the meta-cognitive protocol compiler adaptively adjusts an assembly order of system role instructions and user inputs in the translated instruction stream according to a preset heterogeneous large model attention-sensitive weight matrix;
for a computing power engine sensitive to system instructions, the meta-cognitive protocol compiler writes exclusive mantras into a preset system message slot;
for a computing power engine insensitive to system instructions, the meta-cognitive protocol compiler compiles and encapsulates exclusive mantras into structured tags with specific delimiters and splices the structured tags to a tail of a user request stream.

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The human-machine collaboration system according to Claim 1, characterized in that the Physical Sincerity assertion predicate package specifies boundary constraint conditions including a maximum logical entropy value **H_max**, a forbidden semantic vocabulary list, and upper and lower output Token limits.

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The human-machine collaboration system according to Claim 5, characterized in that after intercepting the raw output streams, the outbound self-consistency assertion interceptor calculates a real-time logical entropy value **H** of the raw output streams based on semantic distribution probabilities and generation probability distributions of vocabularies in the raw output streams; release is permitted if $H \leq$ **H_max** and the raw output streams contain no vocabularies from the forbidden semantic vocabulary list; otherwise, a communication interruption mechanism is forcibly triggered.

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The human-machine collaboration system according to Claim 1, characterized in that when the meta-cognitive protocol compiler dynamically compiles and assembles the exclusive mantra parameter mapping set and the Physical Sincerity assertion predicate package into the translated instruction stream, it further embeds concealed watermark features generated based on the globally unique hash primary key at specific positions of the translated instruction stream; the concealed watermark features are digital watermark features generated by implicitly encoding at least one of control Token sequences, special punctuation encodings, and blank placeholders, such that external monitoring devices can complete protocol property right confirmation verification by sampling and analyzing interaction messages.

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A human-machine collaboration method based on standardized cognitive protocols, executed by a computer processor, characterized by comprising the following steps:

Protocol loading and cascading hash verification: a meta-cognitive protocol compiler loads a standardized cognitive protocol container, performs a cascading hash operation on rule payloads inside the container and the globally unique hash fingerprint of the parent protocol declared in the metadata header of the container, and compares an operation result with the globally unique hash primary key declared by the container to verify integrity;

Engine adaptive compilation: the meta-cognitive protocol compiler identifies a physical identifier of a target computing power engine, dynamically maps and assembles exclusive mantras in the container into a translated instruction stream according to a physical sensitivity matrix corresponding to the engine, and submits the translated instruction stream to the target computing power engine;

and outbound interception audit: an outbound self-consistency assertion interceptor acquires raw text streams returned by the target computing power engine, runs the Physical Sincerity assertion predicate package in the container, and verifies whether characteristic indicators of the raw text streams fall within specified mathematical boundaries; decrypted return is provided to an external user upon passing verification, and output is blocked otherwise.

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The human-machine collaboration method according to Claim 8, characterized in that the specific steps of the cascading hash operation in Step 1 comprise: performing standardized serialization on

the exclusive mantra parameter mapping set and the Physical Sincerity assertion predicate package to generate an ontology data stream, splicing the ontology data stream with the globally unique hash fingerprint of the parent protocol by byte string concatenation, and invoking a hardware-accelerated SHA-256 algorithm to generate a current hash primary key value.

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The human-machine collaboration method according to Claim 8, characterized in that the assembly steps of the translated instruction stream in Step 2 comprise:
parsing parameter placeholders in exclusive mantras, and performing strongly typed binding filling on external input data slots and parameter mapping sets of the exclusive mantras;
and automatically compiling filled mantra streams into system prompt formats required by a specific engine or XML text blocks with delimiters according to detected attention preference weights of the target computing power engine during large model inference.

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The human-machine collaboration method according to Claim 8, characterized in that verifying the characteristic indicators of the raw text streams in Step 3 comprises: calculating a real-time logical entropy value **H** of the raw text streams based on semantic distribution probabilities and generation probability distributions of vocabularies inside the raw text streams, determining whether **H** exceeds the maximum logical entropy value **H_max** set in the assertion predicate package; if exceeded, resetting output and throwing a communication exception signal indicating assertion verification failure to an external system.

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The human-machine collaboration method according to Claim 8, characterized in that the assembly steps of the translated instruction stream further comprise:
when compiling and assembling the translated instruction stream, embedding concealed watermark features generated based on the globally unique hash primary key at specific positions of the translated instruction stream; the concealed watermark features are digital watermark features generated by implicitly encoding at least one of control Token sequences, special punctuation encodings, and blank placeholders; the concealed watermark features are used for completing property right confirmation and infringement verification of the protocol by externally sampling interaction messages.

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A computer-readable storage medium storing a computer program thereon, characterized in that the program, when executed by a processor, implements the method according to any one of Claims 8 to 12.